visStatistics: Automated selection and visualisation of statistical hypothesis tests in R

by Sabine Schilling

Abstract We present **visStatistics**, an R package that automates the selection and visualisation of statistical hypothesis tests between two vectors. Given vectors of class `numeric`, `integer`, or `factor`, the main function `visstat()` determines the appropriate test from the data types, distributional assumptions, and sample size alone, without any manual intervention. Visual outputs — box plots, bar charts, regression lines with confidence bands, mosaic plots, and residual diagnostics — are annotated with test statistics, assumption-check results, and post-hoc comparisons where applicable. The decision logic follows the general linear model framework, assessing normality on pooled model residuals rather than per group, which preserves statistical power. The package is particularly suited for server-side R applications where end users interact exclusively via a web interface, and for time-constrained settings such as statistical consulting or introductory teaching. All tests are applied using their default settings from base R, keeping the dependency footprint minimal and the decision tree transparent.

1 Introduction

Most routine data analyses reduce to a comparatively small set of inferential frameworks: group comparisons, contingency-table analyses, correlation, and simple regression (Sato et al., 2017; Chicco et al., 2025). Selecting the appropriate test within each framework, however, requires evaluating variable types, sample sizes, and distributional assumptions — a process that is error-prone, time-consuming, and a recurring bottleneck in statistical consulting and undergraduate teaching.

Several R packages address parts of this problem. **compareGroups** (Subirana et al., 2014) automates test choice via a per-group Shapiro–Wilk normality test and produces publication-ready descriptive tables, but its primary output is tabular rather than graphical, and assumption diagnostics are not displayed alongside results. **ggstatsplot** (Patil, 2021) produces information-rich **ggplot2**-based (Wickham, 2016) plots with embedded test results, but requires the user to select the appropriate plotting function for each data-type combination (e.g., `ggbetweenstats()` for group comparisons, `ggscatterstats()` for regression). **rstatix** (Kassambara, 2025) provides a pipe-friendly framework for common tests with tidy output, and **ggpubr** (Kassambara, 2026) adds statistical annotations to publication-ready plots; both leave test selection entirely to the user.

visStatistics takes a different approach: a single function `visstat(x, y)` accepts any combination of numeric and categorical vectors and resolves the full decision pipeline — test selection, execution, assumption checking, and visualisation — without any further input. Three design choices distinguish it from the packages above.

First, normality is assessed on the pooled residuals of a linear model `lm(y ~ x)`, following the general linear model (GLM) framework (Searle, 1971), rather than per group. Group-wise testing is statistically less powerful and more likely to reject normality unnecessarily, pushing the analysis to non-parametric methods (Lumley et al., 2002). When heteroscedasticity is subsequently detected, group-wise normality diagnostics are shown additionally, since Welch's methods use group-specific variance estimates.

Second, assumption diagnostics are always displayed, regardless of whether assumptions are met. This enables users to visually override the automated decision in individual cases, and supports the pedagogical goal of making assumption checking an integral part of the analysis rather than an afterthought.

Third, all tests use their default settings from base R functions, keeping the dependency footprint minimal and the decision tree transparent. The package does not wrap external statistical engines; the automated layer is purely in the routing logic.

2 Package overview

visStatistics is available on CRAN. The package accepts input in three equivalent forms:

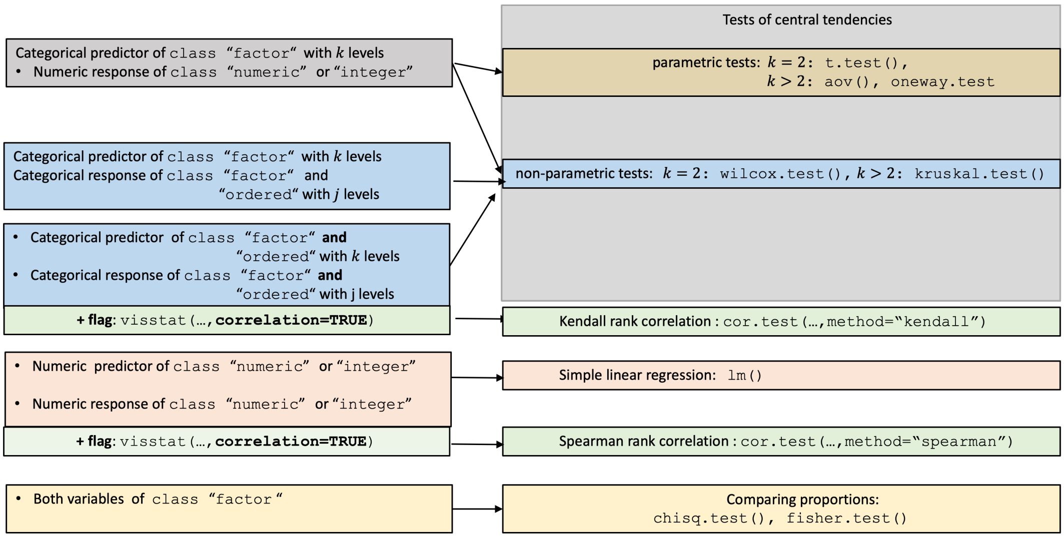

Figure 1: Overview of all implemented tests by input class.

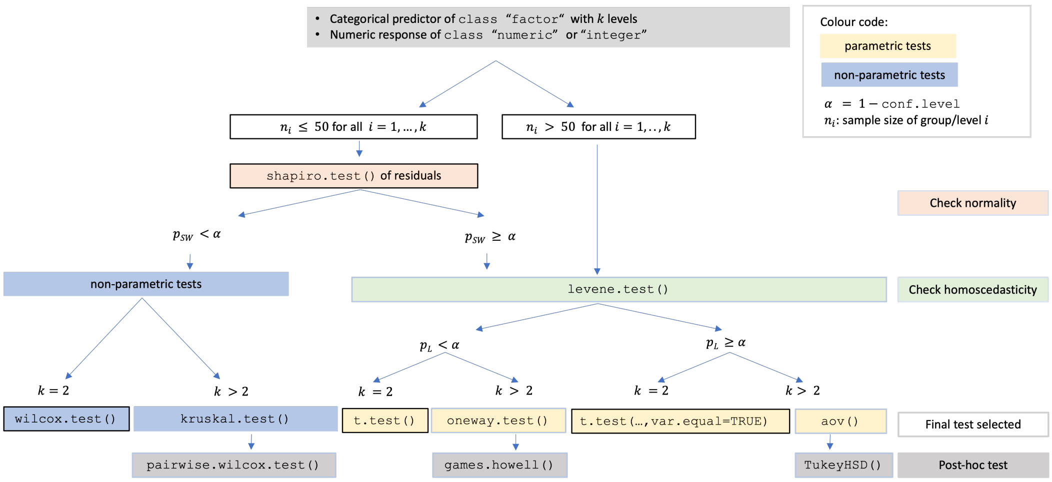

Figure 2: Decision tree for a numeric response and categorical predictor, based on sample size, normality of residuals, and homoscedasticity.

```
# Recommended form:
visstat(x, y)

# Formula interface:
visstat(y ~ x, data = dataframe)

# Backward-compatible form:
visstat(dataframe, "namey", "namex")
```

where x and y are vectors of class "numeric", "integer", or "factor". The function returns an object of class "visstat" with print(), summary(), and plot() S3 methods, and optionally writes graphics to disk via **Cairo** (Urbanek and Horner, 2025).

3 Decision logic

The significance level $\alpha = 1 - \texttt{conf.level}$ is user-controlled (default 0.95). Figure 1 gives a high-level overview of all implemented tests by input class; Figure 2 details the decision path for a numeric response and categorical predictor. Technical definitions of all test statistics are collected in Appendix C.

3.1 Numeric response and categorical predictor: comparing central tendencies

A linear model `lm(y ~ x)` is fitted and assumption diagnostics are always displayed via `vis_lm_assumptions()` (see Appendix A). The Shapiro–Wilk test (Shapiro and Wilk, 1965) is applied to internally studentised residuals from `rstandard()` (Cook and Weisberg, 1982). When all groups contain more than 50 observations, normality is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011).

If normality is rejected ($p_{SW} \leq \alpha$), non-parametric tests are selected: `wilcox.test()` (Mann and Whitney, 1947) for two groups, or `kruskal.test()` (Kruskal and Wallis, 1952) followed by Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979) for more than two groups.

If normality is not rejected, the Levene–Brown–Forsythe test `levene.test()` (Brown and Forsythe, 1974) (see Appendix B) assesses variance homogeneity. For homoscedastic data, Student's `t.test(var.equal = TRUE)` or Fisher's `aov()` (Fisher and Yates, 1990) with `TukeyHSD()` (Hochberg and Tamhane, 1987) is applied. For heteroscedastic data, Welch's `t.test()` (Welch, 1947; Satterthwaite, 1946) or `oneway.test()` (Welch, 1951) with `games.howell()` (Games and Howell, 1976) is applied. When switching to Welch methods, group-wise normality diagnostics are additionally displayed via `vis_group_normality()`, since the pooled `lm()` residuals are no longer the appropriate diagnostic under heteroscedasticity.

3.2 Both variables numeric: regression or rank correlation

By default, `visstat()` fits a simple linear regression via `lm()`, regardless of whether GLM assumptions are met, and always displays assumption diagnostics. When `correlation = TRUE` is set explicitly, Spearman's ρ is computed via `cor.test(..., method = "spearman")`. Because the choice between modelling a directional relationship and measuring a monotone association cannot be automated from data types alone, Spearman correlation is never triggered automatically.

3.3 Both variables categorical: comparing proportions

When both variables are categorical, `visstat()` tests independence using Pearson's χ^2 test (`chisq.test()`) or Fisher's exact test (`fisher.test()`), depending on expected cell counts following Cochran's rule (Cochran, 1954): the χ^2 approximation is used if no expected cell count is less than 1 and no more than 20% of cells have expected counts below 5. Yates' continuity correction (Yates, 1934) is applied by default to 2×2 tables. For general $R \times C$ tables, Pearson's χ^2 result is supplemented with a mosaic plot from **vcd** (Meyer et al., 2006, 2024) with tiles coloured by Pearson residuals.

3.4 Ordered factors

When the response is an ordered factor, `visstat()` converts it to integer level codes and follows the non-parametric path. When both variables are ordered and `correlation = TRUE`, Kendall's τ_b (Kendall, 1945; Agresti, 2010) is computed instead, as it corrects for ties explicitly — unavoidable with few ordinal levels (Xu et al., 2013).

4 Examples

The number of plots returned by `visstat()` depends on the input types and the outcome of assumption checks. Tests that have no separate assumption diagnostics — Kendall's τ_b for two ordered factors — produce a single plot. Most tests produce two panels: an assumption-diagnostic panel and a result panel. When heteroscedasticity is detected and `visstat()` routes to a Welch method, a third panel with group-wise normality diagnostics is added, since pooled-residual diagnostics are no longer appropriate under unequal variances.

4.1 Two groups: Welch's t-test

With binary factor `am` (transmission type, 0 = automatic, 1 = manual) and continuous response `mpg` (fuel efficiency) from the `mtcars` dataset, `visstat()` fits `lm(mpg ~ am)`, applies Shapiro–Wilk to the pooled residuals (not rejected), and subsequently the Levene–Brown–Forsythe test, which detects heteroscedasticity. The routing therefore leads to Welch's t-test rather than Student's t-test. Because the pooled-residual diagnostic is no longer appropriate under heteroscedasticity, `vis_group_normality()` additionally displays per-group normality checks, yielding three panels in total.

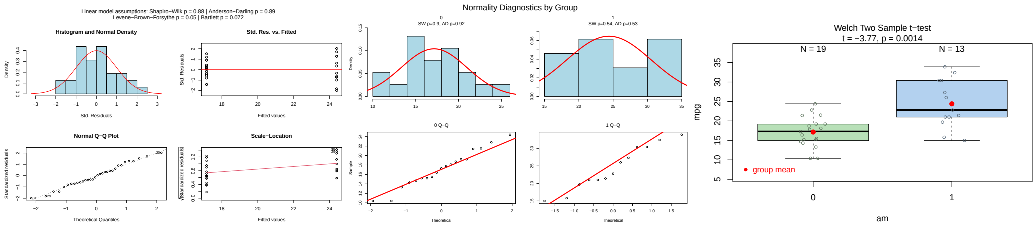

Figure 3: Welch's t-test applied to the mtcars dataset (am vs. mpg). Pooled-residual assumption diagnostics (Shapiro–Wilk, Bartlett, Levene), group-wise normality diagnostics, and box plots of fuel efficiency by transmission type with the Welch t-test result.

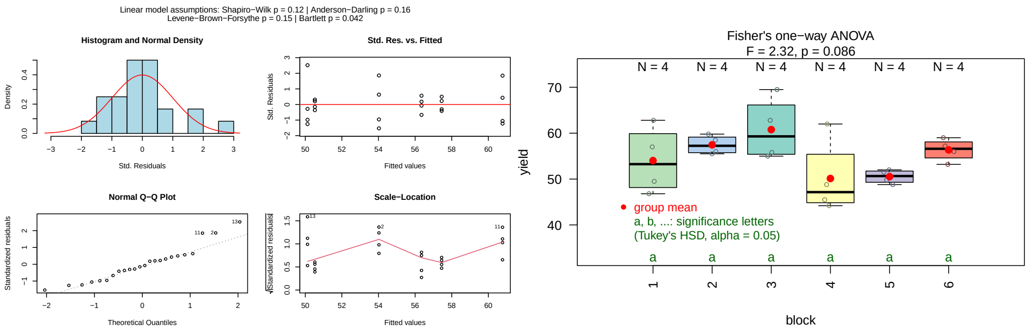

Figure 4: Fisher's one-way ANOVA applied to the npk dataset (block vs. yield). Assumption diagnostics and box plots of pea yield by experimental block; compact letter display (TukeyHSD, $\alpha = 0.05$) shows no significant pairwise differences.

```
mtcars$am <- as.factor(mtcars$am)
t_test_stats <- visstat(mtcars$am, mtcars$mpg)
```

4.2 Multiple groups: one-way ANOVA with Tukey HSD

With factor `block` (experimental block) and continuous response `yield` from the npk pea-growing dataset, Shapiro–Wilk does not reject normality of pooled residuals and the Levene test does not reject homoscedasticity. Fisher's one-way ANOVA is therefore applied, followed by Tukey HSD post-hoc comparisons with a compact letter display.

```
anova_npk <- visstat(npk$block, npk$yield, conf.level = 0.95)
```

4.3 Numeric response and numeric predictor: simple linear regression

With trunk `Girth` and `Volume` from the `trees` dataset, both vectors are numeric, so `visstat()` fits `lm(Volume ~ Girth)` and displays confidence and prediction bands at the user-specified level `conf.level = 0.9`. Assumption diagnostics include the Shapiro–Wilk and Anderson–Darling tests for residual normality and the Breusch–Pagan test for homoscedasticity (see Appendix B).

```
linreg_trees <- visstat(trees$Girth, trees$Volume, conf.level = 0.9)

#> Warning: Statistical assumptions violated:
#> Homoscedasticity violated (Breusch-Pagan p = 0.0178 )
#> Analysis proceeded but interpret results cautiously.

#> RECOMMENDATION: Consider exploring alternatives outside visstat() such as data transformations, generalised li
```

4.4 Two categorical variables: Pearson's χ^2 test

With `Eye` and `Hair` from the `HairEyeColor` dataset, all expected cell counts exceed the Cochran thresholds (Cochran, 1954), so the χ^2 approximation is used. For this 4×4 table, `visstat()` returns both a

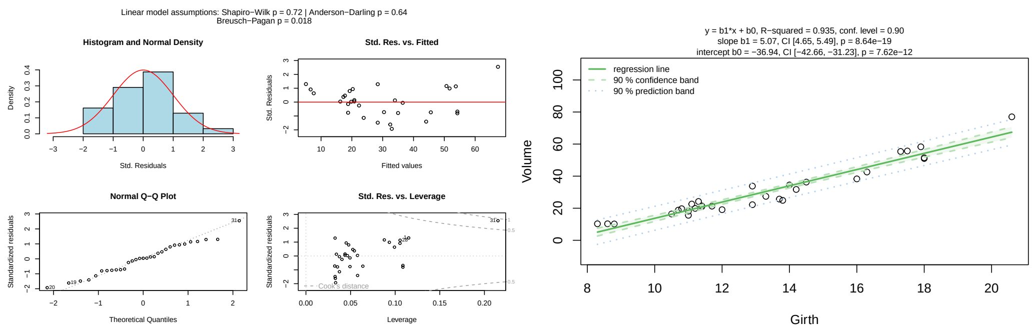

Figure 5: Simple linear regression of `Volume` on `Girth` for the `trees` dataset (`conf.level = 0.9`). Assumption diagnostics (Shapiro–Wilk, Anderson–Darling, Breusch–Pagan) and scatter plot with fitted regression line, 90% confidence band (dark shading), and 90% prediction band (light shading).

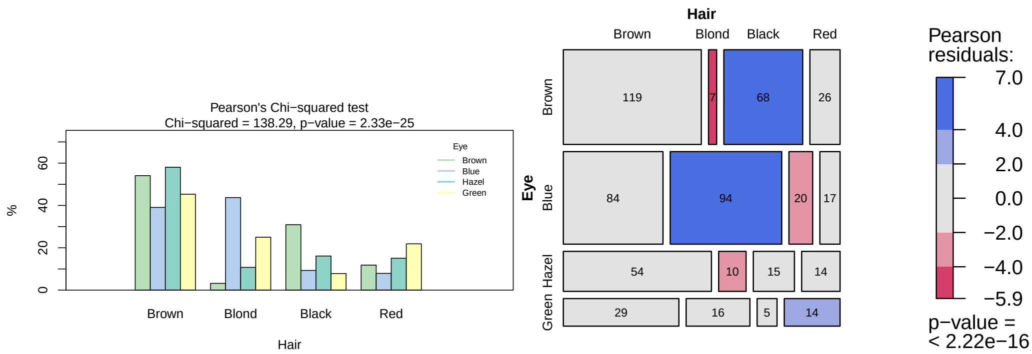

Figure 6: Pearson's χ^2 test applied to the `HairEyeColor` dataset. Grouped bar chart of eye colour by hair colour and mosaic plot with tiles coloured by Pearson residuals (blue: over-represented, red: under-represented).

bar chart of row percentages and a mosaic plot with tiles coloured by Pearson residuals, making the cells driving the association immediately visible.

```
hair_eye_df <- counts_to_cases(as.data.frame(HairEyeColor))
visstat(hair_eye_df$Eye, hair_eye_df$Hair)
```

5 Saving graphical output

All plots can be saved in any format supported by **Cairo** (Urbanek and Horner, 2025) — `"png"`, `"pdf"`, `"svg"`, `"ps"`, `"tiff"` — by supplying the `graphicsoutput` and `plotDirectory` arguments. File names follow the pattern `testname_namey_namex.ext` by default, or a user-supplied `plotName`. The full file paths of generated graphics are stored as the `"plot_paths"` attribute on the returned `"visstat"` object.

6 Limitations

6.1 Scope of automated test selection

All tests are applied with their default settings from base R. As a consequence, paired tests are not supported and interaction terms are not available in the ANOVA path. Only simple linear regression is implemented, as multiple regression cannot be visualised in two dimensions. When regression assumptions are violated, **visStatistics** offers Spearman rank correlation (`correlation = TRUE`) as the sole automated alternative. Alternative methods such as data transformations, generalised linear models, or robust regression are not implemented: each requires user judgment — about the transformation family, the link function, or the estimator — that cannot be automated without substantially expanding the decision tree and increasing the risk of Type I error inflation.

6.2 P-value-based assumption testing

No single test maintains optimal Type I error rates and power across all distributions (Olejnik and Algina, 1987). Normality tests behave poorly at both extremes of the sample-size range: they fail to detect non-normality in small samples and flag negligible departures as significant in large ones (Ghasemi and Zahediasl, 2012; Fagerland, 2012). The $n > 50$ per-group threshold for bypassing the Shapiro–Wilk test is necessarily arbitrary; simulation studies indicate that the required sample size for the central limit theorem to take effect depends on the skewness and kurtosis of the underlying distribution (Fagerland, 2012). Assumption tests provide no information on the nature of the deviation (Shatz, 2024), and combining multiple tests inflates the overall Type I error rate. For these reasons, the diagnostic plots generated by `visstat()` should always be inspected alongside the automated decision; the `conf.level` argument provides a direct handle on the α used throughout.

6.3 Visualisation

The package depends exclusively on base R graphics with no **ggplot2** (Wickham, 2016) dependency, keeping the transitive dependency footprint minimal. For more polished, annotated plots of chosen statistical tests, see **ggstatsplot** (Patil, 2021) or **ggpubr** (Kassambara, 2026).

7 Summary

visStatistics provides a single-function interface to automated statistical test selection, execution, and visualisation for the most commonly used inferential frameworks in two-variable analyses. Its principal contribution is the integration of the full pipeline — data-type detection, distributional assumption checking on GLM residuals, test routing, and annotated graphical output — into a reproducible, server-deployable workflow with a minimal dependency footprint. The package is available on CRAN at `https://CRAN.R-project.org/package=visStatistics` and on GitHub at `https://github.com/shhschilling/visStatistics`.

8 Appendix A: The general linear model

The general linear model (Searle, 1971) provides a unified framework underlying Student's t-test, Fisher's ANOVA, and simple linear regression:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{k-1} x_{i,k-1} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

Student's t-test uses one binary indicator variable x_{i1}; testing $H_0 : \beta_1 = 0$ is equivalent to testing $H_0 : \mu_1 = \mu_2$.

Fisher's ANOVA uses $k - 1$ indicator variables for k groups; testing $H_0 : \beta_1 = \cdots = \beta_{k-1} = 0$ is equivalent to testing equality of all group means.

Simple linear regression uses one continuous predictor; testing $H_0 : \beta_1 = 0$ examines whether a linear relationship exists.

The two-sample case connects t-tests and ANOVA: $t^2 = F$. For two groups, `t.test(var.equal = TRUE)` and `aov()` yield identical p-values, as do `t.test(var.equal = FALSE)` and `oneway.test()`.

9 Appendix B: Tests for homoscedasticity

9.1 The Levene–Brown–Forsythe test `levene.test()`

The test (Brown and Forsythe, 1974) evaluates equality of variances using the absolute deviations from group medians $z_{ij} = |y_{ij} - \tilde{y}_i|$ and applies a one-way ANOVA F-statistic to the z_{ij}. Using the median rather than the mean makes it robust to skewed data (Allingham and Rayner, 2012). The implementation `levene.test()` mimics the default behaviour of `leveneTest()` in **car** (Fox and Weisberg, 2019) and drives the parametric/Welch branching decision.

9.2 Bartlett's test `bartlett.test()`

Bartlett's test (Bartlett, 1937) has greater power than the Brown–Forsythe version when normality holds (Allingham and Rayner, 2012). Its statistic

$$K^2 = \frac{(N-k)\ln s_p^2 - \sum_{i=1}^{k}(n_i - 1)\ln s_i^2}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_i-1} - \frac{1}{N-k}\right)}$$

follows $\chi^2(k-1)$ under the null. Bartlett's test result is shown in the assumption plot but does **not** drive the branching decision; only `levene.test()` does.

9.3 Breusch–Pagan test `bp.test()`

For simple linear regression, group-based variance tests are not applicable. The Breusch–Pagan test (Breusch and Pagan, 1979) assesses whether the variance of the residuals depends on the predictor values and is shown in the regression assumption plot.

10 Appendix C: Test statistics

10.1 Student's t-test and Welch's t-test

The test statistic for Student's t-test is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}},$$

following a t-distribution with $n_1 + n_2 - 2$ degrees of freedom. Welch's t-test relaxes the homoscedasticity assumption:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with degrees of freedom approximated by the Welch–Satterthwaite equation (Welch, 1947; Satterthwaite, 1946). Welch's methods outperform their classical counterparts when variances differ (Moser and Stevens, 1992; Fagerland and Sandvik, 2009; Delacre et al., 2017).

10.2 Wilcoxon rank-sum test

The test statistic is the Mann–Whitney U statistic (Mann and Whitney, 1947):

$$W = U_1 = W_1 - \frac{n_1(n_1+1)}{2},$$

where $W_1 = \sum_{i=1}^{n_1} R(x_{1,i})$ is the rank sum of the first group. An exact p-value is computed for samples of fewer than 50 observations without ties; otherwise a normal approximation with continuity correction is used.

10.3 Fisher's one-way ANOVA and Welch's heteroscedastic ANOVA

Fisher's ANOVA test statistic is

$$F = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)},$$

following an $F(k-1, N-k)$ distribution. Post-hoc comparisons use `TukeyHSD()` (Hochberg and Tamhane, 1987), which controls the family-wise error rate via the studentised range distribution (Abdi, 2007).

Welch's heteroscedastic ANOVA (`oneway.test()`) (Welch, 1951) uses group-specific variance weights and a Satterthwaite-type adjustment for degrees of freedom. Post-hoc comparisons use `games.howell()` (Games and Howell, 1976), which applies separate variance estimates per pair. Compact letter displays are produced by `multcompLetters()` from **multcompView** (Graves et al., 2026).

10.4 Kruskal–Wallis test

When normality is rejected, `kruskal.test()` (Kruskal and Wallis, 1952) tests whether the groups come from the same population based on ranks:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} n_i \left(\bar{R}_i - \bar{R}\right)^2 ,$$

approximately $\chi^2(k-1)$ under the null. Post-hoc comparisons use Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979).

10.5 Simple linear regression and Spearman rank correlation

The regression plot displays the fitted line $\hat{y} = b_0 + b_1 x$ with pointwise confidence and prediction bands at the specified `conf.level`. Assumption diagnostics always include the Shapiro–Wilk and Anderson–Darling (Gross and Ligges, 2015) tests for residual normality and the Breusch–Pagan test `bp.test()` (Breusch and Pagan, 1979) for homoscedasticity (see Appendix B).

When `correlation = TRUE`, Spearman's ρ is computed as Pearson's r applied to the ranks:

$$\rho = r(\mathrm{rank}(x), \mathrm{rank}(y)) .$$

10.6 Pearson's χ^2 test and Fisher's exact test

Let O_{ij} and E_{ij} denote the observed and expected frequencies in row i and column j of an $R \times C$ contingency table. The Pearson residual for cell (i, j) is

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}} .$$

The test statistic of Pearson's χ^2 test (Pearson, 1900) is

$$\chi^2 = \sum_{i,j} r_{ij}^2 ,$$

compared to $\chi^2((R-1)(C-1))$. For general $R \times C$ tables, `visstat()` supplements the bar chart with a mosaic plot in which tiles are coloured by r_{ij} (blue: positive, red: negative), making the cells driving the association immediately visible.

Fisher's exact test (Fisher, 1970) is applied when Cochran's rule (Cochran, 1954) is violated. It evaluates the exact hypergeometric probability of the observed table given the fixed margins.

10.7 Kendall's τ_b

For two ordinal variables with n joint observations, Kendall's τ_b (Kendall, 1945) is

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - n_1)(n_0 - n_2)}} ,$$

where C and D are the numbers of concordant and discordant pairs, $n_0 = n(n-1)/2$, and n_1, n_2 correct for ties in the response and predictor respectively. Kendall's τ_b is preferred over Spearman's ρ for ordinal data with few levels, where ties are common (Agresti, 2010; Xu et al., 2013).

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